Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3900 purchases across various categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

-Rows : 3900
-Columns : 18
-Key Features
-Customer demographics (Age, Gender, Location, Subscription Status)
-Purchase details(Item Purchased, Category, Purchase Amount, Season, Size, Color)
-Shopping behavior (Discount Applies, Promo Code Used, Previous Purchases, Frequency of Purchase ,Review Rating, Shipping Type)
-Missing Data : 37 values in Review Rating column

3. Exploratory Data Analysis Using Python

We began with data preparation and cleaning in python:

- **Data Loading :** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and df.describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Su
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization :** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**
  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days column from purchase data.

- **Data Consistency Check :** Verified if discount_applied and promo_code_used were redundant dropped promo_code_used.

- **Database Integration :** Connected Python script to PostegreSQL and loaded the cleaned DataFrame into the database for SQL analysis

.

4. Exploratory Data Analysis Using Python

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue By Gender :** Compare total revenue generated by male vs female customers.

	gender text	revenue numeric
1	Male	157890
2	Female	75191

2. **High –Spending Discount Users –** Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94

3. **Top 5 Products by Rating:** Found products with the highest average review rating.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

4. **Shipping Type Comparison:** Compare the average Purchase Amount between Standard and Express shipping.

	shipping_type text	Average Amount numeric
1	Standard	58.46
2	Express	60.48

5. **Subscription VS Non-Subscription:** Compare average spend and total revenue across subscription status.

	subscription_status text	Total Customer bigint	Spend numeric	Total Revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

6. **Discount-Dependent Products :** Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation:** Classified customers into New, Returnin, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products Per Category:** Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150

9. **Repeat Buyers & Subscriptions:** Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group:** Calculated total revenue contribution of each age group.

	age_group text	revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI to present insights visually.**

6. Business Recommendations

- **Boost Subscriptions :** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs :** Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy :** Balance sales boosts with margin control.
- **Product Positioning :** Highlight top-rated and best-selling products in campaigns.

THANK YOU